

SQL Keys Notes

Introduction to SQL Keys

Keys in SQL are used to identify records uniquely in a table and to establish relationships between tables.

They are very important in database management systems because they help:

- Maintain data integrity
- Avoid duplicate data
- Create relationships between tables
- Improve searching and indexing

Types of SQL Keys

1. Primary Key
2. Foreign Key
3. Composite Key
4. Candidate Key
5. Alternate Key
6. Super Key
7. Unique Key

Example Tables Used in Notes

Students Table

student_id	roll_no	email	name	department_id
101	CSE01	arun@gmail.com	Arun	1
102	CSE02	priya@gmail.com	Priya	1
103	ECE01	ravi@gmail.com	Ravi	2

Departments Table

department_id	department_name
1	CSE
2	ECE
3	EEE

1. Primary Key

Definition

A **Primary Key** is a column (or set of columns) that uniquely identifies each record in a table.

A primary key:

- Cannot contain NULL values
- Must contain unique values
- There can be only ONE primary key in a table

Syntax

```
CREATE TABLE Students (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(50)  
);
```

Example

```
CREATE TABLE Students (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    age INT  
);
```

Table Example

student_id	name	age
101	Arun	20
102	Priya	19
103	Ravi	21

Here:

- `student_id` uniquely identifies every student.
- No duplicate or NULL values are allowed.

Invalid Example

```
INSERT INTO Students VALUES (101, 'Kiran', 22);
```

This causes an error because `101` already exists.

Important Points

- Uniquely identifies records.
- Automatically creates an index.
- Improves query performance.
- Used for relationships with foreign keys.

2. Foreign Key

Definition

A `Foreign Key` is a column that creates a relationship between two tables.

It refers to the primary key of another table.

Purpose

- Maintains referential integrity

- Prevents invalid data insertion
- Connects related tables

Syntax

```
CREATE TABLE Students (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    department_id INT,  
    FOREIGN KEY (department_id)  
    REFERENCES Departments(department_id)  
);
```

Example

Departments Table

department_id	department_name
1	CSE
2	ECE

Students Table

student_id	name	department_id
101	Arun	1
102	Priya	1
103	Ravi	2

Here:

- `department_id` in Students table is a foreign key.
- It references `department_id` in Departments table.

Invalid Example

```
INSERT INTO Students VALUES (104, 'Meena', 5);
```

Error occurs because department `5` does not exist.

Important Points

- Can contain duplicate values.
 - Can contain NULL values.
 - Multiple foreign keys are allowed in a table.
 - Used to establish table relationships.
-

3. Composite Key

Definition

A `Composite Key` is a key formed using two or more columns together to uniquely identify a record.

Individually, the columns may not uniquely identify rows.

Syntax

```
CREATE TABLE Enrollment (  
    student_id INT,  
    course_id INT,  
    PRIMARY KEY (student_id, course_id)  
);
```

Example

Enrollment Table

student_id	course_id
101	C101
101	C102
102	C101

Here:

- `student_id` alone is not unique.
 - `course_id` alone is not unique.
 - Combination of both becomes unique.
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Important Points

- Uses multiple columns.
 - Common in junction tables.
 - Helps in many-to-many relationships.
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4. Candidate Key

Definition

A `Candidate Key` is a column (or set of columns) that can uniquely identify records.

A table can have multiple candidate keys.

One candidate key is selected as the primary key.

Example

Students Table

student_id	roll_no	email
101	CSE01	arun@gmail.com
102	CSE02	priya@gmail.com

Here:

- `student_id` is unique
- `roll_no` is unique
- `email` is unique

All three are candidate keys.

One of them is selected as the primary key.

Important Points

- Multiple candidate keys can exist.
 - Must contain unique values.
 - Cannot contain NULL values.
-

5. Alternate Key

Definition

An `Alternate Key` is a candidate key that was NOT selected as the primary key.

Example

Suppose:

- `student_id` is chosen as primary key.

Then:

- `roll_no`
- `email`

become alternate keys.

Visualization

Candidate Keys

- `student_id`
- `roll_no`
- `email`

Chosen Primary Key

- `student_id`

Alternate Keys

- `roll_no`
- `email`

Important Points

- Alternate keys are also unique.
 - They can uniquely identify records.
 - They are not selected as primary key.
-

6. Super Key

Definition

A **Super Key** is any combination of columns that can uniquely identify rows in a table.

It may contain extra unnecessary columns.

Example

Students Table

student_id	roll_no	name
101	CSE01	Arun
102	CSE02	Priya

Possible super keys:

- student_id
- roll_no
- student_id + name
- roll_no + name
- student_id + roll_no

All these combinations uniquely identify rows.

Relationship with Candidate Key

- Candidate key = minimal super key.
 - Super key may contain extra columns.
-

Important Points

- Can contain unnecessary columns.
 - Every candidate key is a super key.
 - Every super key is NOT a candidate key.
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7. Unique Key

Definition

A **Unique Key** ensures that all values in a column are unique.

It is similar to a primary key.

Difference:

- Primary key does not allow NULL.
 - Unique key allows one NULL value (in most DBMS).
-

Syntax

```
CREATE TABLE Students (  
    student_id INT PRIMARY KEY,  
    email VARCHAR(100) UNIQUE  
);
```

Example

student_id	email
101	arun@gmail.com
102	priya@gmail.com

Duplicate emails are not allowed.

Invalid Example

```
INSERT INTO Students VALUES (103, 'arun@gmail.com');
```

Error occurs because email must be unique.

Important Points

- Prevents duplicate values.
 - Multiple unique keys allowed.
 - Usually allows one NULL value.
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Difference Between Primary Key and Unique Key

Primary Key	Unique Key
Only one allowed	Multiple allowed
Cannot contain NULL	Can contain NULL
Uniquely identifies rows	Ensures uniqueness
Automatically NOT NULL	NULL allowed

Difference Between Primary Key and Foreign Key

Primary Key	Foreign Key
Uniquely identifies records	References another table
No duplicates	Duplicates allowed
No NULL values	NULL allowed
One per table	Multiple allowed

Difference Between Candidate Key and Super Key

Candidate Key	Super Key
Minimal unique identifier	Can contain extra columns
No unnecessary attributes	May contain unnecessary attributes
Every candidate key is super key	Not every super key is candidate key

Real-World Analogy

Primary Key

Like Aadhaar Number:

- Unique for every person.

Foreign Key

Like Student Department ID:

- Student belongs to a valid department.

Composite Key

Like:

- Train Number + Seat Number

Together identify a ticket uniquely.

Candidate Key

Like:

- Passport Number
- Aadhaar Number
- Driving License Number

All can uniquely identify a person.

Alternate Key

Keys not chosen as primary.

Super Key

Like:

- Aadhaar + Name

Still unique but contains extra data.

Unique Key

Like:

- Email ID in websites.

No duplicates allowed.

Summary Table

Key Type	Unique	NULL Allowed	Multiple Allowed	Purpose
Primary Key	Yes	No	No	Identify records
Foreign Key	No	Yes	Yes	Create relationships
Composite Key	Yes	No	Yes	Combine columns
Candidate Key	Yes	No	Yes	Possible primary keys
Alternate Key	Yes	No	Yes	Candidate keys not selected
Super Key	Yes	Depends	Yes	Uniquely identify rows
Unique Key	Yes	Usually one NULL	Yes	Prevent duplicates

Complete Example

Departments Table

```
CREATE TABLE Departments (  
    department_id INT PRIMARY KEY,  
    department_name VARCHAR(50)  
);
```

Students Table

```
CREATE TABLE Students (  
    student_id INT PRIMARY KEY,  
    roll_no VARCHAR(20) UNIQUE,  
    email VARCHAR(100) UNIQUE,  
    name VARCHAR(50),  
    department_id INT,  
    FOREIGN KEY (department_id)  
    REFERENCES Departments(department_id)  
);
```

Interview Questions

1. Can a table have multiple primary keys?

No. Only one primary key is allowed.

2. Can a foreign key contain NULL values?

Yes.

3. Difference between candidate key and alternate key?

- Candidate key = possible primary key.
 - Alternate key = candidate key not selected.
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4. Is every candidate key a super key?

Yes.

5. Can a unique key contain NULL?

Usually yes (depends on DBMS).

Conclusion

SQL keys are fundamental concepts in database management systems.

They help:

- Maintain uniqueness
- Establish relationships
- Ensure data integrity
- Improve performance

Understanding keys is essential for designing efficient databases and writing reliable SQL queries.